Lecture 4: Stochastic Integration (continued)

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§4 Stochastic Integration (continued)

Wiener–Itô isometry (recap)

We use the following notation for the stochastic integral. For an elementary (step) integrand $\Phi \in \mathcal{E}$, define the (Itô) stochastic integral process

$$I(\Phi)_t \;:=\; \int_0^t \Phi(s)\, dW(s), \qquad t \in [0,T],$$

which is an H-valued continuous square-integrable martingale. Thus I is the map

$$I\colon \mathcal{E} \longrightarrow \mathcal{M}_T^2, \qquad \Phi \longmapsto \big(I(\Phi)_t\big)_{t\in[0,T]}.$$

Recall the Wiener–Itô isometry from Step 2:

$$\big\|I(\Phi)\big\|^2_{\mathcal{M}_T^2} \;=\; \|\Phi\|^2_{\mathcal{E}}, \qquad I(\Phi) \in \mathcal{M}_T^2, \quad \Phi \in \mathcal{E}.$$

Remark 1. Here $\mathcal{E}$ denotes the space of elementary (step) processes from Step 1, equipped with the norm

$$\|\Phi\|^2_{\mathcal{E}} := E\Big(\int_0^T \|\Phi_s \circ \sqrt{Q}\|^2_{L_2(U,H)}\, ds\Big),$$

and $\mathcal{M}_T^2$ is the space of continuous square-integrable H-valued martingales on $[0,T]$. The isometry says that the stochastic integral $I\colon \mathcal{E} \to \mathcal{M}_T^2$ is a **norm-preserving** (isometric) linear map: the L^2-norm of the integral equals the norm of the integrand. This is the starting point for extending I beyond elementary processes by a density/completion argument.

Step 3: Extension to the closure $\overline{\mathcal{E}}$

What is the closure $\overline{\mathcal{E}}$? We want to extend I from the elementary processes $\mathcal{E}$ to a larger space $\overline{\mathcal{E}}$.

Remark 2. Since $I\colon \mathcal{E} \to \mathcal{M}_T^2$ is an isometry, it extends uniquely by continuity to the **closure** of $\mathcal{E}$ in the $\|\cdot\|_{\mathcal{E}}$-norm. The question is: what does this closure look like concretely? The answer involves the **predictable σ-algebra**, which identifies the "right" class of integrands.

Predictable σ-algebra. Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, P)$, where $\mathcal{F}_t$ denotes the σ-algebra of events observable up to time t (the *filtration* of available information).

Define the **predictable σ-algebra** $\mathcal{P}_T$ on $\Omega \times [0,T]$ by

$$\mathcal{P}_T \;:=\; \sigma\Big(\big\{]s,t] \times F_s :\; 0 \le s < t \le T,\; F_s \in \mathcal{F}_s\big\} \;\cup\; \big\{\{0\} \times F_0 :\; F_0 \in \mathcal{F}_0\big\}\Big).$$

Equivalently,

$$\mathcal{P}_T \;=\; \sigma\big(\{(H_s)_{s\in[0,T]} :\; (H_s) \text{ is left-continuous and } (\mathcal{F}_t)\text{-adapted}\}\big).$$

This is called the **predictable σ-algebra**.

Remark 3. The predictable σ-algebra $\mathcal{P}_T$ is generated by sets of the form $]s,t] \times F_s$ with $F_s \in \mathcal{F}_s$ ("stochastic rectangles"), together with $\{0\} \times F_0$. Intuitively, a process is $\mathcal{P}_T$-measurable if and only if it is **left-continuous and adapted** (or, more precisely, a limit of such processes). Predictability is a slightly stronger condition than being merely adapted: it requires the integrand at time t to be determined by information strictly *before* time t, rather than at time t. This is the correct measurability condition for integrands in Itô's stochastic integral, ensuring that the integrand does not "look into the future" of the driving noise.

The closure $\overline{\mathcal{E}}$. Then

$$\overline{\mathcal{E}} \;=\; L^2\big(\Omega_T,\; \mathcal{P}_T,\; P_T;\; L_2^0\big)/\!\sim,$$

where $\sim$ denotes identification up to classes of equivalence, and:

- $\Omega_T := \Omega \times [0,T]$,
- $P_T := P \otimes dt$ (product of the probability measure with Lebesgue measure on $[0,T]$),
- $U_0 := \sqrt{Q}\,U$,
- $L_2^0 := L_2(U_0, H)$ is the space of Hilbert–Schmidt operators from U_0 to H.

Remark 4. The space $\overline{\mathcal{E}} = L^2(\Omega_T, \mathcal{P}_T, P_T; L_2^0)/\!\sim$ consists of all $\mathcal{P}_T$-measurable processes $\Phi\colon \Omega \times [0,T] \to L_2^0$ satisfying

$$\|\Phi\|_{\overline{\mathcal{E}}}^2 \;:=\; E\bigg(\int_0^T \big\|\Phi(s) \circ \sqrt{Q}\,\big\|_{L_2(U,H)}^2\, ds\bigg) \;<\; \infty,$$

with two processes identified if they agree P_T-almost everywhere. The notation $L_2^0 = L_2(U_0, H)$ with $U_0 = \sqrt{Q}\,U$ highlights that we only require $\Phi(s) \circ \sqrt{Q}$ (not $\Phi(s)$ itself) to be Hilbert–Schmidt (when $\Phi(s)$ is represented as an operator on U). This is natural because the Wiener–Itô isometry involves $\|\Phi_s \circ \sqrt{Q}\|_{L_2(U,H)}$: the noise enters through $\sqrt{Q}$, so the integrand need only be "well-behaved" on the range of $\sqrt{Q}$.

Step 4: Localization

As in the finite-dimensional setting, the stochastic integral I can be defined on all

$$\Phi\colon \Omega_T \longrightarrow L_2^0, \qquad \text{adapted, } \mathcal{P}_T\text{-measurable},$$

subject only to the condition

$$P\bigg(\int_0^T \big\|\Phi(s) \circ \sqrt{Q}\,\big\|_{L_2(U,H)}^2\, ds \;<\; \infty\bigg) \;=\; 1.$$

Remark 5. In Steps 2–3, the stochastic integral was defined for integrands satisfying the L^2-**condition**: $E\big(\int_0^T \|\Phi(s)\circ\sqrt{Q}\|^2_{L_2(U,H)}\,ds\big) < \infty$ (finiteness *in expectation*). The localization step weakens this to a **pathwise (almost sure)** condition: $\int_0^T \|\Phi(s)\circ\sqrt{Q}\|^2_{L_2(U,H)}\,ds < \infty$ holds P-a.s., without requiring the expectation to be finite. This extension is achieved by a **stopping-time argument**: one "truncates" Φ to make it L^2-integrable, defines the integral on each truncation, and verifies consistency. The resulting integral is a **local martingale** (rather than a true martingale), exactly as in the scalar Itô theory. This is the largest natural domain for the stochastic integral.

Properties of the stochastic integral

(i) Linearity (composition with bounded operators). For $L \in L(H, \widetilde{H})$ (a bounded linear operator between Hilbert spaces, with $\widetilde{H}$ another Hilbert space), we have

$$L\left(\int_0^t \Phi(s)\,dW(s)\right) \;=\; \int_0^t L\circ\Phi(s)\,dW(s).$$

Remark 6. This property says that the stochastic integral **commutes with bounded linear operators**. Concretely: applying a bounded linear transformation $L\colon H \to \widetilde{H}$ to the integral $\int_0^t \Phi(s)\,dW(s) \in H$ is the same as integrating the composed process $L\circ\Phi(s)\colon U \to \widetilde{H}$. The proof follows directly from the definition for elementary processes (where L can be pulled through the finite sum) and extends to general integrands by the density/isometry argument. This is the infinite-dimensional analogue of the scalar property $c\cdot\int_0^t \phi(s)\,dB_s = \int_0^t c\,\phi(s)\,dB_s$ for a constant $c \in \mathbb{R}$.

(ii′) Inner product with a fixed vector. Let $h \in H$ be fixed. Then

$$\left\langle h,\, \int_0^t \Phi(s)\,dW(s)\right\rangle_H \;=\; \int_0^t \underbrace{h^*\circ\Phi(s)}_{=\,\widetilde{\Phi}_h(s)}\,dW_s,$$

where $\widetilde{\Phi}_h(s) := h^*\circ\Phi(s) \in L(U,\mathbb{R})$, with $h^* = \langle h,\,\cdot\,\rangle_H\colon H \to \mathbb{R}$ denoting the linear functional induced by $h \in H$ via the Riesz representation.

Remark 7. Property (ii′) says that the fixed linear functional $\langle h,\,\cdot\,\rangle_H$ can be "pulled inside" the stochastic integral. The operator $\widetilde{\Phi}_h(s) = \langle h,\,\Phi(s)(\cdot)\rangle_H : U \to \mathbb{R}$ maps $u \in U$ to the scalar $\langle h,\,\Phi(s)\,u\rangle_H$, so the right-hand side is a **real-valued** stochastic integral. Equivalently, for each fixed h, the scalar process $t \mapsto \langle h, \int_0^t \Phi(s)\,dW(s)\rangle_H$ is exactly the Itô integral of the scalar integrand $u \mapsto \langle h, \Phi(s)u\rangle_H$:

$$\left\langle h, \int_0^t \Phi(s)\,dW(s)\right\rangle_H = \int_0^t h^*\Phi(s)\,dW(s).$$

This follows first for elementary processes by linearity of the inner product and then for general Φ by the L^2-isometry/approximation argument. The identification of h with the functional $h^* = \langle h,\cdot\rangle_H$ is precisely the Riesz representation theorem in the Hilbert space H. For a time-varying (possibly random) adapted process $\hat{f}(t)$, the displayed identity is *not* valid in general; one must use an Itô product rule (integration by parts), which produces additional terms involving $d\hat{f}$ and quadratic covariation.

(iii) Burkholder–Davis–Gundy (BDG) inequalities. For every $p \geq 1$ there exists a constant C_p (depending only on p) such that for every integrand $\Phi \in \overline{\mathcal{E}}$,

$$E\Big(\sup_{0\leq t\leq T} \Big\| \int_0^t \Phi(s)\, dW(s)\Big\|_H^p\Big) \leq C_p\, E\Big(\Big(\int_0^T \|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^2\, ds\Big)^{p/2}\Big).$$

Remark 8. The BDG inequalities are the natural maximal estimates for martingales. They follow from the scalar BDG applied to the canonical coordinate Brownian motions together with the Hilbert–Schmidt representation of the noise; alternatively one can reduce to the finite-dimensional case via projections and then pass to the limit using the isometry. BDG is essential for pathwise regularity estimates and for controlling the supremum norm of stochastic integrals.

(iv) Quadratic variation. Let $M_t := I(\Phi)_t = \int_0^t \Phi(s)\, dW(s)$. Then the predictable quadratic variation of M is given by

$$\langle M\rangle_t \;=\; \int_0^t \|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^2\, ds, \qquad t\in[0,T].$$

Moreover the process $\|M_t\|_H^2 - \langle M\rangle_t$ is a real-valued martingale.

Remark 9. The identity for the quadratic variation is the infinite-dimensional analogue of $\langle \int_0^\cdot \phi_s\, dB_s\rangle_t = \int_0^t \phi_s^2\, ds$. It is proved first for step integrands by direct computation of cross terms, then extended to general Φ by density and the Wiener–Itô isometry. The process $\|M_t\|_H^2 - \langle M\rangle_t$ is a martingale by the Itô formula for $\|M_t\|_H^2$ (or equivalently by applying the scalar isometry to each coordinate).

(v) Regularity of the stochastic integral (Sobolev–Slobodetskii). Fix $\alpha \in (0, \frac{1}{2})$. Define the fractional Sobolev (Slobodetskii) seminorm for an H-valued process M by

$$[M]_{W^{\alpha,2}([0,T];H)}^2 := \int_0^T\int_0^T \frac{\|M_t - M_s\|_H^2}{|t-s|^{1+2\alpha}}\, ds\, dt.$$

If $\Phi \in \overline{\mathcal{E}}$ satisfies the L^2-condition, then $M = I(\Phi)$ belongs to the space $L^2(\Omega; W^{\alpha,2}([0,T];H))$ and one has the estimate

$$E\big([M]_{W^{\alpha,2}}^2\big) \leq C(\alpha)\, E\Big(\int_0^T \|\Phi(s)\circ\sqrt{Q}\|_{L_2(U,H)}^2\, ds\Big) \;<\; \infty.$$

Remark 10. Recall that

$$W^{\alpha,2}([0,T];H) = \big\{f \in L^2(0,T;H) : [f]_{W^{\alpha,2}([0,T];H)} < \infty\big\}.$$

Let $M = I(\Phi)$. By symmetry,

$$E\big([M]_{W^{\alpha,2}}^2\big) = 2\int_0^T\int_t^T \frac{E\big(\|M_s - M_t\|_H^2\big)}{(s-t)^{1+2\alpha}}\, ds\, dt.$$

For $0 \leq t < s \leq T$ we have $M_s - M_t = \int_t^s \Phi(r)\, dW(r)$, hence by the Itô isometry,

$$E\big(\|M_s - M_t\|_H^2\big) = E\Big(\int_t^s \|\Phi(r)\circ\sqrt{Q}\|_{L_2(U,H)}^2\, dr\Big).$$

Insert this and apply Tonelli:

$$
\begin{aligned}
E\big([M]^2_{W^{\alpha,2}}\big) &= 2\,E\int_0^T\int_t^T\int_t^s \frac{\|\Phi(r)\circ\sqrt{Q}\|^2_{L_2(U,H)}}{(s-t)^{1+2\alpha}}\,dr\,ds\,dt \\
&= 2\,E\int_0^T \|\Phi(r)\circ\sqrt{Q}\|^2_{L_2(U,H)}\Big(\int_0^r\int_r^T (s-t)^{-1-2\alpha}\,ds\,dt\Big)\,dr.
\end{aligned}
$$

For each fixed $r\in[0,T]$, the kernel is finite when $\alpha<\frac{1}{2}$ and satisfies

$$
\int_0^r\int_r^T (s-t)^{-1-2\alpha}\,ds\,dt \le C(\alpha)<\infty,
$$

so we obtain

$$
E\big([M]^2_{W^{\alpha,2}}\big) \le C(\alpha)\;E\Big(\int_0^T \|\Phi(r)\circ\sqrt{Q}\|^2_{L_2(U,H)}\,dr\Big).
$$

Together with $E\int_0^T \|M_t\|_H^2\,dt<\infty$ (from the Itô isometry), this yields $M\in L^2(\Omega;W^{\alpha,2}([0,T];H))$.

Cylindrical Wiener process

The Q-Wiener process $W_t=\sum_{n\ge1}\sqrt{\lambda_n}\,\beta_n(t)\,e_n$ converges in $L^2(\Omega;U)$ provided $\mathrm{tr}(Q)=\sum_n\lambda_n<\infty$. In this case the stochastic integral $\int_0^t\Phi(s)\,dW_s$ is well-defined for predictable Φ such that

$$
E\int_0^T \|\Phi(s)\|^2_{L_2(U_0,H)}\,ds \;=\; E\int_0^T \|\Phi(s)\circ\sqrt{Q}\|^2_{L_2(U,H)}\,ds \;<\; \infty.
$$

When Q is not trace class one can still make sense of the "noise" by working with the **cylindrical Wiener process**

$$
W_t \;=\; \sum_{n\ge1}\beta_n(t)\,e_n,
$$

which does *not* converge in U. In that case the stochastic integral is defined via

$$
\int_0^t \Phi(s)\,dW_s \;:=\; \sum_{n\ge1}\int_0^t \Phi(s)e_n\,d\beta_n(s),
$$

and it is well-defined as an H-valued continuous local martingale whenever Φ is predictable and

$$
\int_0^T \|\Phi(s)\|^2_{L_2(U,H)}\,ds<\infty \quad P\text{-a.s.}
$$

(in particular, if the above integral is finite in expectation).

Remark 11. The cylindrical Wiener process corresponds formally to $Q=\mathrm{Id}$ (all eigenvalues equal to 1). In this case $\mathrm{tr}(Q)=\infty$, so the series $\sum_n\beta_n(t)\,e_n$ does not converge in U. However, the stochastic integral against it is still meaningful: since $\sqrt{Q}=\mathrm{Id}$ and $U_0=U$, one needs $\Phi(s)$ itself to be Hilbert–Schmidt from U to H, i.e. $\Phi(s)\in L_2(U,H)$. Cylindrical Wiener processes arise naturally in practice (e.g. space–time white noise on L^2), and the entire stochastic integration theory developed above carries over to this setting without change, since the integral was always defined in terms of $\Phi\circ\sqrt{Q}$.

Stochastic evolution equations on Hilbert spaces

Throughout this section we fix two separable (real) Hilbert spaces U, H and a Q-Wiener process $(W_t)_{t\geq 0}$ with respect to a filtration $(\mathcal{F}_t)_{t\geq 0}$ on U (or, when Q is not trace class, a cylindrical Wiener process).

The (infinite-dimensional) SPDE / stochastic evolution equation. Consider the stochastic evolution equation

$$dX_t \;=\; \big[A\,X_t + B(X_t)\big]\,dt \;+\; C(X_t)\,dW_t \quad \text{in } H, \qquad X_0 = \xi, \tag{4.1}$$

Remark 12. In *(4.1)*, the unknown is an H-valued stochastic process $X = (X_t)_{t\in[0,T]}$. The term $A\,X_t$ is the *linear drift* (typically generated by a differential operator, e.g. a Laplacian-type operator), while $B(X_t)$ is a (possibly nonlinear) *drift/nonlinear forcing* term. The term $C(X_t)\,dW_t$ is the *diffusion/noise* term: for each state $x \in H$, $C(x)$ is an operator-valued coefficient (from the noise space into H), and W is the driving Q-Wiener (or cylindrical Wiener) process. Finally, ξ is the initial condition at time $t = 0$. Thus *(4.1)* is the infinite-dimensional analogue of a finite-dimensional Itô SDE. In this setting one often calls *(4.1)* an SPDE (or stochastic evolution equation), with A encoding the spatial evolution operator and C describing how the noise enters the system.

subject to the following assumptions.

Assumptions.

(A1) $(A, D(A))$ generates a C_0-semigroup $(T_t)_{t\geq 0}$ on H.

(A2) $B\colon H \to H$ is $\mathcal{B}(H)$-measurable.

(A3) $C\colon H \to L_2(U_0, H)$ is **strongly continuous**, i.e. for each fixed $u \in U_0$, the map

$$x \;\longmapsto\; C(x)\,u, \qquad H \to H,$$

is continuous. Here $U_0 := \sqrt{Q}\,U$ denotes the *effective noise space* associated with the covariance operator Q (the range of $\sqrt{Q}$, equipped with the natural Hilbert structure used in the stochastic integral).

(A4) ξ is H-valued and $\mathcal{F}_0$-measurable.

Remark 13. **(A1)** says that the linear part A of the drift generates a strongly continuous (i.e. C_0-) semigroup. This is the standard framework for abstract parabolic and hyperbolic evolution equations (e.g. the Laplacian on L^2 generates the heat semigroup).

(A2) is a minimal measurability assumption on the nonlinear drift B. In applications one typically imposes Lipschitz or local Lipschitz conditions on B to obtain existence and uniqueness; here we state the weakest structural requirement. In particular, if X is an H-valued adapted/measurable process, then $B(X_t)$ is again H-valued measurable for each t, so the drift term can be interpreted as a Bochner integral (under suitable integrability).

(A3) requires the diffusion coefficient C to be *strongly* continuous (continuity of $C(x)\,u$ in x for each fixed u), rather than norm-continuous. Strong continuity is strictly weaker than norm (operator-norm) continuity and suffices for the measurability properties needed in the stochastic integral.

(A4) is the usual initial-condition assumption: ξ is an H-valued random variable known at time 0.

C_0-semigroups

A family of bounded linear operators $(T_t)_{t\geq 0}$ on H is called a C_0**-semigroup** if:

1. $T_0 = \mathrm{Id}$,
2. $T_t \circ T_s = T_{t+s}$ for all $t, s \geq 0$ (semigroup property),
3. $t \mapsto T_t\, u$ is continuous for every $u \in H$ (strong continuity).

The **generator** of the semigroup is the (generally unbounded) operator A defined by

$$A\,h \;:=\; \lim_{t\to 0^+} \frac{T_t\, h - T_0\, h}{t} \;=\; \lim_{t\to 0^+} \frac{T_t\, h - h}{t},$$

with domain $D(A) := \{h \in H : \text{the limit above exists in } H\}$.

Remark 14. The generator A encodes the "infinitesimal" behaviour of the semigroup: $T_t \approx \mathrm{Id} + t\,A$ for small t. This explains the terminology *infinitesimal generator*: for $h \in D(A)$, the limit $Ah = \lim_{t\to 0^+}(T_t h - h)/t$ is the time derivative of $t \mapsto T_t h$ at $t = 0$, so A describes the first-order effect of an infinitesimal time step. For the heat equation on $H = L^2(\mathbb{R}^d)$, $A = \frac{1}{2}\Delta$ and T_t is the heat semigroup (convolution with the Gaussian kernel). More precisely, for $t > 0$ let

$$G_t(x) := (2\pi t)^{-d/2} \exp\bigl(-|x|^2/(2t)\bigr), \qquad x \in \mathbb{R}^d,$$

then for $u_0 \in L^2(\mathbb{R}^d)$ one has

$$(T_t u_0)(x) = (G_t * u_0)(x) := \int_{\mathbb{R}^d} G_t(x-y)\, u_0(y)\, dy.$$

The semigroup property $T_t T_s = T_{t+s}$ corresponds to the Gaussian convolution identity $G_t * G_s = G_{t+s}$. In general A is an *unbounded* operator (defined only on a dense domain $D(A) \subset H$), and the Hille–Yosida theorem characterises exactly which operators generate C_0-semigroups.

The semigroup property $T_t \circ T_s = T_{t+s}$ is the operator analogue of $e^{at}\, e^{as} = e^{a(t+s)}$; indeed when A is bounded one simply has $T_t = e^{tA}$.

Notions of solution

We discuss three standard notions of solutions for (**??**).

Mild solution. An H-valued predictable process $(X_t)_{t\in[0,T]}$ is called a **mild solution** to (**??**) if for all $t \in [0, T]$,

$$X_t \;=\; T_t\, \xi \;+ \int_0^t T_{t-s}\, B\bigl(X_s\bigr)\, ds \;+ \int_0^t T_{t-s}\, C\bigl(X_s\bigr)\, dW_s, \qquad P\text{-a.s.} \tag{4.2}$$

Remark 15. The mild formulation replaces the (potentially ill-defined) term $A\,X_t$ by the semigroup T_{t-s} (variation of constants / Duhamel principle). Intuitively, T_{t-s} is the linear flow generated by A from time s to time t: it propagates a contribution created at time s forward to time t under the linear dynamics. Thus the forcing terms $B(X_s)\, ds$ and $C(X_s)\, dW_s$ injected at time s appear at time t as $T_{t-s}B(X_s)\, ds$ and $T_{t-s}C(X_s)\, dW_s$. This is crucial because solutions typically do not lie in $D(A)$, so the expression $A\,X_t$ may not make sense pointwise. The two integrals in *(4.2)* must be well-defined: the Bochner integral for $T_{t-s}B(X_s)$ and the stochastic integral for $T_{t-s}C(X_s)$.

Strong (analytical) solution. An H-valued predictable process (X_t) is called a **(analytically) strong solution** if

- $X_t \in D(A)$ for a.e. t (and P-a.s.),
- $\int_0^T \|A\,X_s\|_H\,ds < \infty$ and the integrals below exist,

and for all $t \in [0,T]$,

$$X_t \;=\; \xi \;+\int_0^t \big[A\,X_s + B(X_s)\big]\,ds \;+\int_0^t C(X_s)\,dW_s, \qquad P\text{-a.s.}$$

Remark 16. The term *strong solution* is used in two different senses in the stochastic evolution equation literature: (i) strong in the probabilistic sense (solution adapted to the given Wiener process), and (ii) strong in the analytical sense (solution possesses enough spatial regularity so that $A\,X_t$ is meaningful). Here we mean the **analytical** notion.

Weak (analytical) solution. An H-valued predictable process (X_t) is called an **(analytically) weak solution** if for every test vector $\varphi \in D(A^*)$ and all $t \in [0,T]$,

$$\langle X_t, \varphi\rangle_H \;=\; \langle \xi, \varphi\rangle_H \;+\int_0^t \langle X_s, A^*\varphi\rangle_H\,ds \;+\int_0^t \langle B(X_s), \varphi\rangle_H\,ds \;+\int_0^t \langle \varphi, C(X_s)\,dW_s\rangle_H, \qquad P\text{-a.s.}$$

where $(A^*, D(A^*))$ denotes the adjoint operator of A.

Remark 17. In the weak formulation, one tests the equation against $\varphi \in D(A^*)$ so that $\langle A\,X_s, \varphi\rangle_H$ can be interpreted as $\langle X_s, A^*\varphi\rangle_H$ without requiring $X_s \in D(A)$. This parallels the classical PDE notion of weak solution.

Existence and uniqueness of mild solutions

We impose the following conditions on B and C.

(H.1) Lipschitz condition. There exists $L > 0$ such that for all $x, y \in H$,

$$\|B(x) - B(y)\|_H^2 \;+\|\big(C(x) - C(y)\big)\circ\sqrt{Q}\|_{L_2(U,H)}^2 \;\le\; L\,\|x-y\|_H^2.$$

(H.2) Linear growth. There exists $M > 0$ such that for all $x \in H$,

$$\|B(x)\|_H^2 \;+\|C(x)\circ\sqrt{Q}\|_{L_2(U,H)}^2 \;\le\; M\,\big(1 + \|x\|_H^2\big).$$

Theorem 1 (Existence and uniqueness of a mild solution)**.** Assume **(A1)**–**(A4)** and **(H.1)**–**(H.2)**. In addition, assume that $E\|\xi\|_H^p < \infty$ for some $p \ge 2$ (equivalently $\xi \in L^p(\Omega; H)$). Then there exists a unique mild solution $(X_t)_{t\in[0,T]}$ of (**??**). Moreover,

$$P\bigg(\int_0^T \|X_s\|_H^2\,ds < \infty\bigg) = 1.$$

Remark 18. The proof is typically based on a fixed-point argument (Picard iteration) on the space of predictable processes, using the Wiener–Itô isometry and BDG inequalities to control the stochastic convolution $\int_0^t T_{t-s}C(X_s)\,dW_s$. The Lipschitz assumption **(H.1)** yields contraction (locally in time, then global by iteration), while **(H.2)** ensures integrability. The assumption $E\|\xi\|_H^p < \infty$ is an *integrability hypothesis* on the initial condition, needed because we work in the L^p-based space H_p and must control the term $T_t\xi$ in L^p. It is satisfied, for instance, if ξ is deterministic, bounded, or more generally $\xi \in L^p(\Omega; H)$.

Proof via Banach's fixed point theorem (details)

Fix $p \geq 2$ and define the Banach space

$$H_p := \Big\{ Y : [0,T]\times\Omega \to H : \ Y \text{ is predictable and } \|Y\|_{H_p} < \infty \Big\}, \qquad \|Y\|_{H_p} := \sup_{t\in[0,T]} \big(E\|Y_t\|_H^p\big)^{1/p}.$$

For $Y \in H_p$, define the map K by

$$(KY)(t) := T_t\,\xi + \int_0^t T_{t-s}\,B\big(Y_s\big)\,ds + \int_0^t T_{t-s}\,C\big(Y_s\big)\,dW_s, \qquad t \in [0,T]. \tag{1}$$

Write

$$(KY)(t) = T_t\xi + K_1(Y)(t) + K_2(Y)(t)$$

with

$$K_1(Y)(t) := \int_0^t T_{t-s}B(Y_s)\,ds, \qquad K_2(Y)(t) := \int_0^t T_{t-s}C(Y_s)\,dW_s.$$

Let $M_T := \sup_{r\in[0,T]} \|T_r\|_{L(H)} < \infty$.

Step 1: K maps H_p into itself. Let $Y \in H_p$. Since Y is predictable and B, C are measurable, the processes $B(Y)$ and $C(Y)$ are predictable, and therefore KY is predictable as well. It remains to show that $\sup_{t\in[0,T]} E\|KY(t)\|_H^p < \infty$.

First, we use the elementary inequality

$$\|a+b+c\|_H^p \leq 3^{p-1}\big(\|a\|_H^p + \|b\|_H^p + \|c\|_H^p\big), \qquad a,b,c \in H, \tag{2}$$

which follows from Jensen's inequality applied to the convex function $u \mapsto \|u\|_H^p$. Thus, for each $t \in [0,T]$,

$$E\|KY(t)\|_H^p \leq 3^{p-1}\Big(E\|T_t\xi\|_H^p + E\|K_1(Y)(t)\|_H^p + E\|K_2(Y)(t)\|_H^p\Big).$$

Drift term. By $\|T_{t-s}\|_{L(H)} \leq M_T$,

$$\|K_1(Y)(t)\|_H = \Big\|\int_0^t T_{t-s}B(Y_s)\,ds\Big\|_H \leq \int_0^t \|T_{t-s}\|_{L(H)}\,\|B(Y_s)\|_H\,ds \leq M_T \int_0^t \|B(Y_s)\|_H\,ds.$$

For $p \geq 1$, Hölder's inequality gives

$$\Big(\int_0^t f(s)\,ds\Big)^p \leq t^{p-1}\int_0^t f(s)^p\,ds, \qquad f \geq 0,$$

and therefore,

$$E\|K_1(Y)(t)\|_H^p \leq M_T^p\,t^{p-1}\int_0^t E\|B(Y_s)\|_H^p\,ds. \tag{3}$$

Since the integrand is nonnegative, Tonelli yields

$$E\int_0^t \|B(Y_s)\|_H^p\,ds = \int_0^t E\|B(Y_s)\|_H^p\,ds.$$

Moreover, by **(H.2)** and $(1+\|x\|_H^2)^{p/2} \leq c_p(1+\|x\|_H^p)$,

$$E\|B(Y_s)\|_H^p \leq C(p,M)\big(1+E\|Y_s\|_H^p\big) \leq C(p,M)\big(1+\|Y\|_{H_p}^p\big).$$

In particular, combining with (3) we obtain the simplified estimate

$$E\|K_1(Y)(t)\|_H^p \leq C(p,M,M_T)\,t^p\,\big(1+\|Y\|_{H_p}^p\big) \leq C(p,M,M_T)\,T^p\,\big(1+\|Y\|_{H_p}^p\big).$$

Diffusion term. Fix $t \in [0,T]$ and consider the H-valued martingale

$$N_r := \int_0^r T_{t-s} C(Y_s)\, dW_s, \qquad r \in [0,t].$$

Then $K_2(Y)(t) = N_t$, and BDG (in fixed time form, obtained from the maximal inequality) gives

$$E\|K_2(Y)(t)\|_H^p \le C_p\, E\Big(\int_0^t \|T_{t-s} C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^2\, ds\Big)^{p/2}.$$

Using $\|T_{t-s}\|_{L(H)} \le M_T$ and $\|T_{t-s} S\|_{L_2(U,H)} \le \|T_{t-s}\|_{L(H)}\, \|S\|_{L_2(U,H)}$, we obtain

$$E\|K_2(Y)(t)\|_H^p \le C_p\, M_T^p\, E\Big(\int_0^t \|C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^2\, ds\Big)^{p/2}.$$

Since $p/2 \ge 1$, the same Hölder estimate (with exponent $p/2$) yields

$$\Big(\int_0^t g(s)\, ds\Big)^{p/2} \le t^{\frac{p}{2}-1} \int_0^t g(s)^{p/2}\, ds, \qquad g \ge 0,$$

and with $g(s) = \|C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^2$ we get

$$E\|K_2(Y)(t)\|_H^p \le C_p\, M_T^p\, t^{\frac{p}{2}-1} \int_0^t E\|C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^p\, ds. \tag{4}$$

Since the integrand is nonnegative, Tonelli gives

$$E \int_0^t \|C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^p\, ds = \int_0^t E\|C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^p\, ds.$$

Moreover, by **(H.2)** and $(1 + \|x\|_H^2)^{p/2} \le c_p(1 + \|x\|_H^p)$,

$$E\|C(Y_s) \circ \sqrt{Q}\|_{L_2(U,H)}^p \le C(p,M)\big(1 + E\|Y_s\|_H^p\big) \le C(p,M)\big(1 + \|Y\|_{H_p}^p\big).$$

In particular, combining with (4) we obtain the simplified estimate

$$E\|K_2(Y)(t)\|_H^p \le C(p,M,M_T)\, t^{p/2}\, \big(1 + \|Y\|_{H_p}^p\big) \le C(p,M,M_T)\, T^{p/2}\, \big(1 + \|Y\|_{H_p}^p\big).$$

Equivalently, for fixed $T > 0$ one may write

$$E\|K_2(Y)(t)\|_H^p \le M(T)\, T^{(p+1)/2}\, \big(1 + \|Y\|_{H_p}^p\big),$$

for a constant $M(T)$ depending on (p, M, M_T, T) (absorbing $T^{-1/2}$ into the constant).

Combining the bounds for $T_t\xi$, $K_1(Y)(t)$, and $K_2(Y)(t)$ and taking $\sup_{t\in[0,T]}$, we obtain

$$\sup_{t\in[0,T]} E\|KY(t)\|_H^p \le C\Big(E\|\xi\|_H^p + 1 + \|Y\|_{H_p}^p\Big),$$

for a constant $C = C(p,T,M,M_T)$ (absorbing the factors T^p and $T^{p/2}$). Hence $KY \in H_p$ whenever $Y \in H_p$.

Step 2: Contraction estimate (small time). Let $Y_1, Y_2 \in H_p$ and $t \in [0,T]$. The semigroup term cancels, so

$$(KY_1)(t) - (KY_2)(t) = \big(K_1(Y_1)(t) - K_1(Y_2)(t)\big) + \big(K_2(Y_1)(t) - K_2(Y_2)(t)\big).$$

We will use Jensen's inequality (applied to the convex function $u \mapsto \|u\|_H^p$):

$$\|u+v\|_H^p \le 2^{p-1}\big(\|u\|_H^p + \|v\|_H^p\big), \qquad u, v \in H. \tag{5}$$

Drift difference. By definition,

$$K_1(Y_1)(t) - K_1(Y_2)(t) = \int_0^t T_{t-s}\big(B(Y_{1,s}) - B(Y_{2,s})\big)\, ds.$$

Thus, using $\|T_{t-s}\| \le M_T$, Hölder in time and Tonelli as in Step 1,

$$E\|K_1(Y_1)(t) - K_1(Y_2)(t)\|_H^p \le M_T^p\, t^{p-1} \int_0^t E\|B(Y_{1,s}) - B(Y_{2,s})\|_H^p\, ds.$$

From **(H.1)** we have $\|B(x) - B(y)\|_H^2 \le L\|x-y\|_H^2$; raising to $p/2$ yields

$$\|B(x) - B(y)\|_H^p \le L^{p/2}\|x-y\|_H^p.$$

Hence, for each s,

$$E\|B(Y_{1,s}) - B(Y_{2,s})\|_H^p \le L^{p/2}\, E\|Y_{1,s} - Y_{2,s}\|_H^p \le L^{p/2}\, \|Y_1 - Y_2\|_{H_p}^p,$$

and therefore

$$E\|K_1(Y_1)(t) - K_1(Y_2)(t)\|_H^p \le M_T^p\, L^{p/2}\, t^p\, \|Y_1 - Y_2\|_{H_p}^p. \tag{6}$$

Diffusion difference. Similarly,

$$K_2(Y_1)(t) - K_2(Y_2)(t) = \int_0^t T_{t-s}\big(C(Y_{1,s}) - C(Y_{2,s})\big)\, dW_s.$$

By BDG (fixed time form), $\|T_{t-s}\| \le M_T$ and the same Hölder estimate in time,

$$E\|K_2(Y_1)(t) - K_2(Y_2)(t)\|_H^p \le C_p\, M_T^p\, t^{\frac{p}{2}-1} \int_0^t E\|\big(C(Y_{1,s}) - C(Y_{2,s})\big) \circ \sqrt{Q}\|_{L_2(U,H)}^p\, ds.$$

Using **(H.1)** in the form $\|((C(x) - C(y)) \circ \sqrt{Q})\|_{L_2(U,H)}^p \le L^{p/2}\|x-y\|_H^p$, we obtain

$$E\|K_2(Y_1)(t) - K_2(Y_2)(t)\|_H^p \le C_p\, M_T^p\, L^{p/2}\, t^{p/2}\, \|Y_1 - Y_2\|_{H_p}^p. \tag{7}$$

Combine and take $\sup_{t\in[0,T]}$. By (5), (6), and (7), for each $t \in [0,T]$,

$$E\|(KY_1)(t) - (KY_2)(t)\|_H^p \le \widetilde{C}\left(t^p + t^{p/2}\right) \|Y_1 - Y_2\|_{H_p}^p,$$

for $\widetilde{C} = \widetilde{C}(p, L, M_T)$. Taking $\sup_{t\in[0,T]}$ yields

$$\|KY_1 - KY_2\|_{H_p}^p \le \widetilde{C}\left(T^p + T^{p/2}\right) \|Y_1 - Y_2\|_{H_p}^p.$$

Choosing $T_0 > 0$ such that $\widetilde{C}(T_0^p + T_0^{p/2}) < 1$, the map K is a strict contraction on H_p over $[0, T_0]$.

Step 3: Local fixed point and global solution. Since H_p is complete and K is a strict contraction on $[0, T_0]$, Banach's fixed point theorem yields a unique fixed point $X \in H_p$ such that $X = KX$ on $[0, T_0]$. By definition of K, this is exactly the mild formulation *(4.2)* on $[0, T_0]$. In particular, for the given initial condition ξ, there is a unique mild solution on the time interval $[0, T_0]$.

To extend to $[0, T]$, one concatenates the local solutions. Set $t_0 := 0$ and $t_{n+1} := \min\{t_n + T_0, T\}$. Assuming X is constructed on $[0, t_n]$, define the next segment on $[t_n, t_{n+1}]$ as the fixed point of the corresponding map with initial data X_{t_n}, namely

$$X_t = T_{t-t_n} X_{t_n} + \int_{t_n}^{t} T_{t-s} B(X_s)\, ds + \int_{t_n}^{t} T_{t-s} C(X_s)\, dW_s, \qquad t \in [t_n, t_{n+1}].$$

The contraction estimate depends only on the interval length (at most T_0) and the constants in **(H.1)**–**(H.2)** and M_T, so the same argument applies on each interval $[t_n, t_{n+1}]$. Hence there is a unique mild solution on each such interval given the left-endpoint value X_{t_n}, and by induction this yields existence and uniqueness on the full horizon $[0, T]$.

Step 4: Moment bound and continuity. We briefly indicate the ingredients for the a priori moment bound. Starting from the mild formulation $X = KX$ and taking $\sup_{t\in[0,T]}$, one estimates: (i) $\sup_t \|T_t \xi\|_H \le M_T \|\xi\|_H$ by boundedness of the semigroup, (ii) the drift term by Hölder in time together with **(H.2)**, and (iii) the stochastic term by a maximal BDG/Doob estimate together with $\|T_{t-s}\| \le M_T$ and **(H.2)**. This yields a bound of the form

$$E\Big(\sup_{t\in[0,T]} \|X_t\|_H^p\Big) \le C_{p,T}\left(1 + E\|\xi\|_H^p\right),$$

with $C_{p,T}$ depending on (p, T, L, M, M_T). In particular, since $p \ge 2$,

$$\begin{aligned} E\int_0^T \|X_s\|_H^2\, ds &= \int_0^T E\|X_s\|_H^2\, ds \\ &\le \int_0^T \left(E\|X_s\|_H^p\right)^{2/p} ds \\ &\le T\Big(\sup_{t\in[0,T]} E\|X_t\|_H^p\Big)^{2/p} \\ &\le T\Big(E\sup_{t\in[0,T]} \|X_t\|_H^p\Big)^{2/p} < \infty, \end{aligned}$$

where we used Tonelli, the inequality $\|Z\|_{L^2} \le \|Z\|_{L^p}$ for $p \ge 2$, and the moment bound above. In particular,

$$T\Big(E\sup_{t\in[0,T]} \|X_t\|_H^p\Big)^{2/p} \le T\left(C_{p,T}\big(1 + E\|\xi\|_H^p\big)\right)^{2/p} < \infty.$$

Since the random variable $\int_0^T \|X_s\|_H^2\, ds$ is nonnegative, finiteness of its expectation implies

$$P\Big(\int_0^T \|X_s\|_H^2\, ds < \infty\Big) = 1,$$

which is the integrability statement in the theorem (indeed, if it were infinite with positive probability, then its expectation would be infinite). Finally, X admits a continuous modification (see the remark below).

Remark 19. **Why we care about a continuous modification.** The process X is initially constructed as a family of random variables $(X_t)_{t\in[0,T]}$, each defined only up to changes on a null set that may depend on t. A *continuous modification* is another version $\widetilde{X}$ such that $P(\widetilde{X}_t = X_t \text{ for all } t) = 1$ and, in addition, the sample paths $t \mapsto \widetilde{X}_t(\omega)$ are continuous. We care because many arguments in SPDE theory (and similarly in SDE theory) are *pathwise* and require an actual trajectory with good time-regularity: it lets us talk meaningfully about quantities such as $\sup_{t\in[0,T]} \|X_t(\omega)\|_H$, sample-path Hölder regularity, and hitting/exit times (which are stopping times when paths are continuous). It also matches the usual definition of a solution as an H-valued process with continuous paths.

In the mild formula (1), each term has a continuous modification: $t \mapsto T_t\xi$ is continuous by strong continuity of (T_t), the drift is a time-integral (hence continuous in t), and the stochastic convolution admits a continuous modification (e.g. via BDG for increments together with the regularity estimates from the integration theory above).

Remark 20. **Intuition for Steps 1–4.**

- Step 1 (define one Picard step): think of K as the map "plug a guess Y into the right-hand side of the mild equation and compute the next iterate." To make this meaningful we must check two things: (i) measurability/predictability, so the stochastic integral is defined, and (ii) integrability, so the drift and diffusion terms do not blow up in L^p. The growth bound **(H.2)** is exactly what turns $\|B(Y_s)\|$ and $\|C(Y_s)\circ\sqrt{Q}\|$ into something controlled by $1+\|Y_s\|$, allowing the estimates in H_p.

- Step 2 (make the Picard step contract): compare two guesses Y_1, Y_2 and estimate the size of $(KY_1)(t) - (KY_2)(t)$. The Lipschitz bound **(H.1)** says that changing the input by $\|Y_1 - Y_2\|$ changes the coefficients by at most a constant times $\|Y_1 - Y_2\|$. What makes this a *contraction* is small time: the drift term involves an integral in time (so it brings in a factor of order t), while the stochastic term behaves like a Brownian integral (typical size of order $\sqrt{t}$). After taking p-moments, these become factors t^p and $t^{p/2}$, so for T small enough the overall multiplicative factor is < 1.

- Step 3 (cover the full time interval): Banach's theorem gives one unique fixed point on $[0, T_0]$, i.e. one unique solution path segment starting from ξ. To go beyond T_0 you do not create independent solutions; you *restart* with initial data equal to the endpoint value already produced, X_{T_0}, and apply the same local argument on $[T_0, 2T_0]$, and so on. This produces one single global path on $[0, T]$.

- Step 4 (turn estimates into path properties): once you have L^p bounds on $\sup_{t\le T} \|X_t\|$, you can deduce $E\int_0^T \|X_s\|_H^2 ds < \infty$, hence $\int_0^T \|X_s\|_H^2 ds < \infty$ holds P-a.s. The mild formula also shows that X has a continuous modification: the semigroup term is continuous, the drift is a time-integral, and the stochastic convolution admits a continuous modification.